

Deepak Mallampati

APPLIED AI ENGINEER - AGENTIC & AUTONOMOUS SYSTEMS

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US-based, F-1 OPT - open to Austin relocation & sponsorship

PROFESSIONAL SUMMARY

AI-native engineer with 4+ years building and deploying production agentic and LLM systems end to end. Architected a conductor-subagent agentic framework that decomposes complex objectives into autonomous, context-scoped subagents executing multi-step workflows - cutting token consumption 40%+ while sustaining task accuracy. Deep engineering foundation across APIs, retrieval, and observability (12 microservices serving 100,000+ requests/day at p95 800ms), with a track record of shipping independently - including a live Chrome Web Store product across 29+ sites. Brings regulated-domain rigor (HIPAA, audit logging, privacy-preserving workflows) that transfers directly to insurance and safety-critical constraints.

CORE COMPETENCIES

Agentic Systems & Orchestration • Autonomous Multi-Step Workflows • LLM Application Engineering • RAG & Retrieval • API & Microservice Integration • MLOps & Observability • Streaming & Data Pipelines • End-to-End Productionization

Generative AI & Agents: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | Multi-Agent Systems | RAG Architecture | Prompt Engineering | Fine-tuning (LoRA/QLoRA/PEFT) | Hugging Face Transformers

Retrieval & Eval: FAISS, Pinecone, Weaviate | Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching | RAGAS, BERTScore, Drift Detection

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda, EC2/S3), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, W&B | Monitoring & Observability

Data & Streaming: Kafka | Spark, Databricks | PostgreSQL, MongoDB, Redis | Pandas, NumPy | Data Lineage & Governance

Languages & Frameworks: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large healthcare corpora, combining dense retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%** - directly applicable to autonomous agents operating in production.
- Established an **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to catch regressions before release.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing while holding SLA targets.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by 27% and informing platform model-selection.
- Engineered safeguards and content controls for policy-compliant AI responses, reducing unsafe outputs by 42%; delivered LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a **privacy-preserving ML pipeline** on clinical data (k-anonymity, l-diversity, differential privacy), cutting re-identification risk from 3.38% to 0.32% while retaining 99% of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 with a curated instruction dataset for controllable long-form generation; authored an IEEE-format paper.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Trained regression/classification models for equipment-failure prediction and anomaly detection on operational sensor and telemetry data, building end-to-end model-ready pipelines.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving 89% F1 on custom NER tasks.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - a solo, production, end-to-end build with real users.

career-ops

Autonomous, AI-driven pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - agentic orchestration applied to a real workflow.

Story Director LLM

Fine-tuned model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

EDUCATION

M.S. in Computer Science, Kent State University, OH

2025 | GPA 3.70/4.0